

Week 7

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1. Matrix Multiplication

This section shows a different way to denote matrix multiplication. Consider the following vectors

$$u = \begin{bmatrix} a \\ b \end{bmatrix}, v = \begin{bmatrix} c \\ d \end{bmatrix}$$

The following holds true.

$$\begin{aligned} uu^T + vv^T &= \begin{bmatrix} a & c \\ b & d \end{bmatrix} \times \begin{bmatrix} a & b \\ c & d \end{bmatrix} \\ \left(\begin{bmatrix} a \\ b \end{bmatrix} \times \begin{bmatrix} a & b \end{bmatrix} \right) + \left(\begin{bmatrix} c \\ d \end{bmatrix} \times \begin{bmatrix} c & d \end{bmatrix} \right) &= \begin{bmatrix} a^2 + c^2 & ab + cd \\ ab + cd & b^2 + d^2 \end{bmatrix} \\ \begin{bmatrix} a^2 & ab \\ ab & b^2 \end{bmatrix} + \begin{bmatrix} c^2 & cd \\ cd & d^2 \end{bmatrix} &= \begin{bmatrix} a^2 + c^2 & ab + cd \\ ab + cd & b^2 + d^2 \end{bmatrix} \\ \begin{bmatrix} a^2 + c^2 & ab + cd \\ ab + cd & b^2 + d^2 \end{bmatrix} &= \begin{bmatrix} a^2 + c^2 & ab + cd \\ ab + cd & b^2 + d^2 \end{bmatrix} \end{aligned}$$

Consider n data points $x_1, \dots, x_n$. These are column vectors. Matrix X is given as follows :

$$X = \begin{bmatrix} - & x_1^T & - \\ - & x_2^T & - \\ & \vdots & \\ - & x_n^T & - \end{bmatrix}$$

Each row in matrix X is a data point. $X^T X$ can be denoted by

$$X^T X = \sum_{i=1}^n x_i x_i^T = (x_1 x_1^T) + \dots + (x_n x_n^T)$$

2. An Optimization Problem

Let A be an $m \times n$ matrix and let $\text{rank}(A) = k$.

Suppose, among all the unit vectors in $\mathbb{R}^n$, we want to find that vector $\hat{x} \in \mathbb{R}^n$ that maximizes $\|Ax\|^2$.
Formally,

$$\hat{x} = \arg \max_{\substack{x, s.t., \\ \|x\|=1}} [\|Ax\|^2]$$

$s.t.$, denotes '**such that**'.

$A^T A$ is an $n \times n$ symmetric matrix. $A^T A$ is diagonalizable.

$$A^T A = Q D Q^T$$

where,

$$Q = \begin{bmatrix} | & | & \dots & | \\ v_1 & v_2 & \dots & v_n \\ | & | & \dots & | \end{bmatrix}, D = \begin{bmatrix} \lambda_1 & 0 & \dots & 0 \\ 0 & \lambda_2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & \lambda_n \end{bmatrix}$$

Q is an orthogonal matrix. The diagonals of D are arranged such that

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$$

We know that the eigenvalues of $A^T A$ are greater than or equal to 0. The rank of A is k . As a result, $\text{Rank}(A^T A) = k$. Which means, $\lambda_1, \dots, \lambda_k$ are the non-zero eigenvalues of $A^T A$ and $v_1, \dots, v_k$ are the corresponding eigenvectors.

$$\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k > 0$$

Let us expand $\|Ax\|^2$

$$\begin{aligned} \|Ax\|^2 &= (Ax)^T (Ax) \\ &= x^T A^T A x \\ &= x^T Q D Q^T x \end{aligned}$$

Let $Qx = y$, where $y = [y_1 \ y_2 \ \dots \ y_n]^T$. Then,

$$\begin{aligned} \|Ax\|^2 &= x^T Q D Q^T x \\ &= (Qx)^T D (Qx) \\ &= y^T D y \\ &= \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2 \end{aligned}$$

Since $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_n$, we get

$$\begin{aligned} \lambda_1 y_1^2 + \lambda_1 y_2^2 + \dots + \lambda_1 y_n^2 &\geq \lambda_1 y_1^2 + \lambda_2 y_2^2 + \dots + \lambda_n y_n^2 \\ \lambda_1 (y_1^2 + y_2^2 + \dots + y_n^2) &\geq \|Ax\|^2 \\ \lambda_1 \|y\|^2 &\geq \|Ax\|^2 \end{aligned}$$

The maximum value that Ax can take is $\lambda_1 \|y\|^2$. Note, $\|y\| = \|x\|$. This is because, $Qx = y$ and Q is orthogonal matrix.

$$\|Ax\|^2 \leq \lambda_1 \|x\|^2$$

If $\|x\|^2 = 1$, then

$$\|Ax\|^2 \leq \lambda_1$$

Recall, the eigenvector corresponding to λ_1 is v_1 . Also, $\|v_1\| = 1$. If $x = v_1$, then

$$\begin{aligned} \|Av_1\|^2 &= v_1^T A^T A v_1 \\ &= v_1^T (\lambda_1 v_1) \\ &= \lambda_1 (v_1^T v_1) \\ &= \lambda_1 \end{aligned}$$

Among all the unit vectors $x \in \mathbb{R}^n$, $x = v_1$ is the vector that maximizes $\|Ax\|^2$ and the maximum value that $\|Ax\|^2$ can take is λ_1 .

Suppose, among all the unit vectors in $\mathbb{R}^n$ that are orthogonal to v_1 , we want to find the vector which maximizes $\|Ax\|^2$. Formally,

$$\hat{x} = \arg \max_{\substack{x, s.t., \\ \|x\|=1 \ \& \ x^T v_1=0}} [\|Ax\|^2]$$

$$\begin{aligned} \|Ax\|^2 &= (Ax)^T (Ax) \\ &= x^T A^T A x \end{aligned}$$

$$= x^T Q D Q^T x$$

Let $Qx = z$, where $z = \begin{bmatrix} z_1 & z_2 & \cdots & z_n \end{bmatrix}^T$. Then,

$$\begin{aligned} \|Ax\|^2 &= x^T Q D Q^T x \\ &= (Qx)^T D (Qx) \\ &= z^T D z \\ &= \lambda_1 z_1^2 + \lambda_2 z_2^2 + \dots + \lambda_n z_n^2 \end{aligned}$$

$Qx = z$ implies $x = Q^T z$. The solution vector x should satisfy $v_1^T x = 0$.

$$\begin{aligned} v_1^T x &= 0 \\ v_1^T Q z &= 0 \\ v_1^T (z_1 v_1 + z_2 v_2 + \dots + z_n v_n) &= 0 \\ z_1 (v_1^T v_1) + z_2 (v_1^T v_2) + \dots + z_n (v_1^T v_n) &= 0 \\ z_1 \|v_1\|^2 + z_2(0) + \dots + z_n(0) &= 0 \\ z_1 \|v_1\|^2 &= 0 \\ z_1 &= 0 \end{aligned}$$

Recall, $\|v_1\| = 1$. If $\|Ax\|^2$ has to be maximum for some $x = Q^T z$, then $z_1 = 0$.

$$\begin{aligned} \|Ax\|^2 &= \lambda_1 z_1^2 + \lambda_2 z_2^2 + \dots + \lambda_n z_n^2 \\ &= \lambda_1(0) + \lambda_2 z_2^2 + \dots + \lambda_n z_n^2 \\ &= \lambda_2 z_2^2 + \dots + \lambda_n z_n^2 \end{aligned}$$

Since $\lambda_2 \geq \dots \geq \lambda_n$, we get

$$\begin{aligned} \lambda_2 z_2^2 + \lambda_2 z_3^2 + \dots + \lambda_2 z_n^2 &\geq \lambda_2 z_2^2 + \dots + \lambda_n z_n^2 \\ \lambda_2 (z_2^2 + z_3^2 + \dots + z_n^2) &\geq \|Ax\|^2 \\ \lambda_2 \|z\|^2 &\geq \|Ax\|^2 \\ \lambda_2 \|x\|^2 &\geq \|Ax\|^2 \end{aligned}$$

Note, $z = \begin{bmatrix} z_1 & z_2 & \cdots & z_n \end{bmatrix}^T$. Since $z_1 = 0$, $\|z\|^2 = z_2^2 + z_3^2 + \dots + z_n^2$. Also, $\|x\| = \|z\|$ as $Qx = z$ and Q is an orthogonal matrix.

If $\|x\|^2 = 1$

$$\begin{aligned} \|Ax\|^2 &\leq \lambda_2 \|x\|^2 \\ &\leq \lambda_2 \end{aligned}$$

The eigenvector corresponding to λ_2 is v_2 . Also, $\|v_2\| = 1$ and $v_1^T v_2 = 0$. If $x = v_2$, then

$$\begin{aligned}\|Av_2\|^2 &= v_2^T A^T A v_2 \\ &= v_2^T (\lambda_2 v_2) \\ &= \lambda_2 (v_2^T v_2) \\ &= \lambda_2\end{aligned}$$

Among all the unit vectors $x \in \mathbb{R}^n$ that are orthogonal to v_1 , $x = v_2$ is the vector that maximizes $\|Ax\|^2$ and the maximum value that $\|Ax\|^2$ can take is λ_2 .

The rank of $A^T A$ is r . We can repeat the above steps again and again for k times.

Summary

- Among all the unit vectors in $\mathbb{R}^n$, $x = v_1$, which is the eigenvector of $A^T A$ corresponding to the largest eigenvalue of $A^T A$, maximizes $\|Ax\|^2$
 - Among all the unit vectors in $\mathbb{R}^n$ that are orthogonal to v_1 , $x = v_2$, which is the eigenvector of $A^T A$ corresponding to the 2nd largest eigenvalue of $A^T A$, maximizes $\|Ax\|^2$
 - Among all the unit vectors in $\mathbb{R}^n$ that are orthogonal to $\{v_1, \dots, v_{k-1}\}$, $x = v_k$, which is the eigenvector of $A^T A$ corresponding to the k^{th} largest eigenvalue of $A^T A$, maximizes $\|Ax\|^2$
-

3. Covariance Matrix

$\{x_1, \dots, x_n\}$ are n data points in $\mathbb{R}^d$, represented as column vectors. The mean of these data points is 0. Consider matrix X .

$$X = \begin{bmatrix} - & x_1^T & - \\ - & x_2^T & - \\ & \vdots & \\ - & x_n^T & - \end{bmatrix}, \quad X = \begin{bmatrix} | & | & & | \\ f_1 & f_2 & \cdots & f_d \\ | & | & & | \end{bmatrix}$$

Each row in X is a data point and each column in X is a feature vector. The data points are elements in $\mathbb{R}^d$, therefore, there are d feature vectors. The matrix

$$C = \frac{1}{n} \sum_{i=1}^n x_i x_i^T = \frac{1}{n} X^T X$$

is called the **covariance** matrix.

In general, for data points $\{x_1, \dots, x_n\}$, with $\bar{x}$ as the mean of the data points, the covariance matrix, is

given by

$$C = \frac{1}{n} \sum_{i=1}^n (x_i - \bar{x})(x_i - \bar{x})^T$$

As we have mentioned earlier, the mean of our dataset is 0, therefore, $C = \frac{1}{n} \sum_{i=1}^n x_i x_i^T$

Let us expand C

$$\begin{aligned} C &= \frac{1}{n} X^T X \\ &= \frac{1}{n} \begin{bmatrix} - & f_1^T & - \\ - & f_2^T & - \\ & \vdots & \\ - & f_d^T & - \end{bmatrix} \begin{bmatrix} | & | & \cdots & | \\ f_1 & f_2 & \cdots & f_d \\ | & | & & | \end{bmatrix} \\ &= \frac{1}{n} \begin{bmatrix} f_1^T f_1 & f_1^T f_2 & \cdots & f_1^T f_d \\ f_2^T f_1 & f_2^T f_2 & \cdots & f_2^T f_d \\ \vdots & \vdots & \ddots & \vdots \\ f_d^T f_1 & f_d^T f_2 & \cdots & f_d^T f_d \end{bmatrix} \\ &= \begin{bmatrix} \text{Var}(f_1) & \text{Cov}(f_1, f_2) & \cdots & \text{Cov}(f_1, f_d) \\ \text{Cov}(f_2, f_1) & \text{Var}(f_2) & \cdots & \text{Cov}(f_2, f_d) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(f_d, f_1) & \text{Cov}(f_d, f_2) & \cdots & \text{Var}(f_d) \end{bmatrix} \end{aligned}$$

The element in the i^{th} row and j^{th} column is the covariance of feature f_i and feature f_j .

$$\text{Cov}(f_i, f_j) = \frac{f_i^T f_j}{n}$$

If $i = j$, then it becomes the variance of feature f_i .

Properties

- C is a symmetric matrix.
- C is a positive semi-definite matrix. That is, all the eigenvalues of C are greater than or equal to 0.
- C is orthogonally diagonalizable

4. PCA Motivation

Consider the following dataset. The mean of this data set is 0.

$$D = \{x_1, \dots, x_n\}, D \subset \mathbb{R}^d$$

How many real numbers do we need to store the dataset? We have n data points in $\mathbb{R}^d$, therefore, we need $n \times d$ real numbers to store the dataset.

Can we do better? Yes, if D lies in a k dimensional subspace in $\mathbb{R}^d$, $k < d$. Suppose, the following set is a basis set for the k dimensional subspace which contains the dataset.

$$B = \{\beta_1, \dots, \beta_k\}, B \subset \mathbb{R}^d$$

Each data point x_i is a linear combination of the vectors in B . Here, $a_{(i,j)}$ is the coefficient of β_j in the expansion of x_i .

$$x_i = \alpha_{(i,1)}\beta_1 + \alpha_{(i,2)}\beta_2 + \dots + \alpha_{(i,k)}\beta_k$$

For each data point, we need k scalars. For n data points, we need $n \times k$ scalars. Also, we need to store the vectors in B . There are k data points in B . We need $k \times d$ real numbers. Overall, we need to store $(n \times k) + (k \times d) = k(n + d)$ real numbers.

Example

Suppose we have 1000 data points in $\mathbb{R}^{100}$. We would have to store 100000 numbers. If the data points lie in $k = 10$ dimensional subspace in $\mathbb{R}^{100}$, then

$$\begin{aligned} k(n + d) &= 10(1000 + 100) \\ &= 11000 \end{aligned}$$

we can store 11000 real numbers.

Choice for the Basis

To reduce the number of real numbers required to store the dataset, we can technically choose any basis set which spans the k dimensional subspace in $\mathbb{R}^d$ containing the dataset.

Consider the following.

$$X = \begin{bmatrix} - & x_1^T & - \\ - & x_2^T & - \\ & \vdots & \\ - & x_n^T & - \end{bmatrix}$$

Let $C = \frac{1}{n}X^T X$ be the covariance matrix. The data points lie in a k dimensional subspace in $\mathbb{R}^d$. Therefore, $\text{rank}(X) = k$. Also, $\text{rank}(C) = k$

We can find k eigenvalues for C that are greater than 0. They are :

$$\lambda_1 \geq \lambda_2 \geq \dots \lambda_k > 0$$

Let v_i be the eigenvector of C corresponding to λ_i and let $\|v_i\| = 1$. Consider the following set

$$\{v_1, \dots, v_k\}$$

Since C is a symmetric matrix, we can choose the above set as an orthonormal set.

Claim : $D = \text{Span}(\{v_1, \dots, v_k\})$.

Proof :

$$\begin{aligned} C v_i &= \lambda_i v_i \\ \frac{1}{n} X^T X v_i &= \lambda_i v_i \\ X^T X v_i &= n \lambda_i v_i \end{aligned}$$

Here, $X v_i$ is a column vector multiplying X^T to get $n \lambda_i v_i$. When we multiply a matrix with a column vector, the resultant is a linear combination of the columns of the matrix. Therefore, $n \lambda_i v_i$ is a linear combination of the columns of X^T .

$n \lambda_i v_i$ is in the column space of X^T . If a vector is an element of a vector space, then any scalar multiple of that vector is also an element of the vector space. Therefore, v_i is an element of the column space of X^T .

The set $\{v_1, \dots, v_k\}$ is contained in the column space of X^T . Every column in X^T is a data point. Which means, the set $\{v_1, \dots, v_k\}$ is contained in the vector space spanned by all the data points in D .

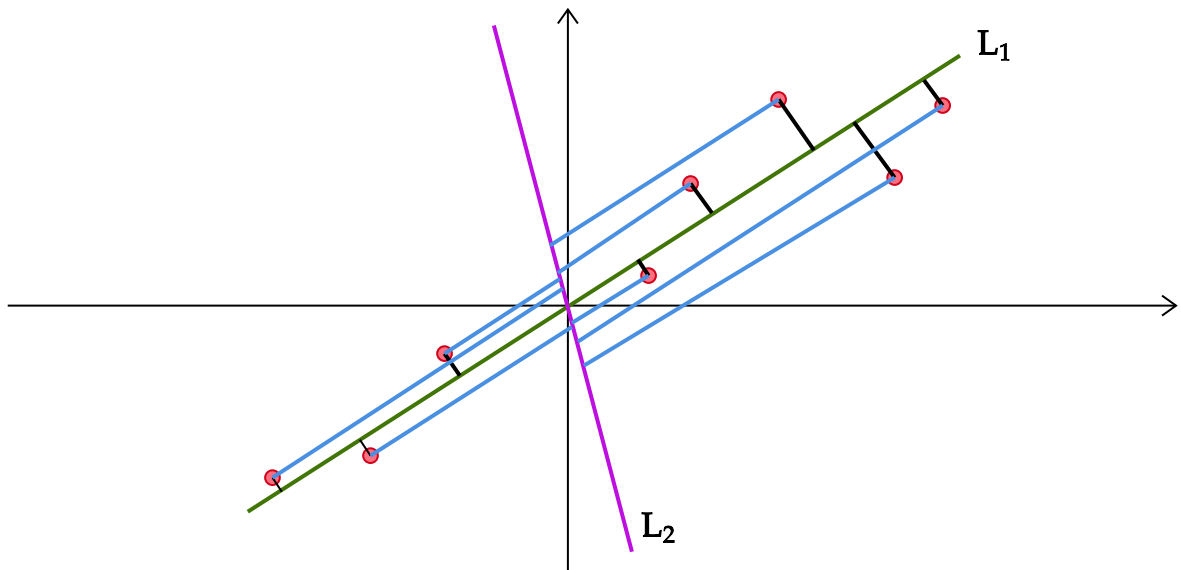
The dataset is contained in a k dimensional subspace of $\mathbb{R}^d$. Similarly, $\text{span}(\{v_1, \dots, v_k\})$ is also a k dimensional subspace in $\mathbb{R}^d$. The $\{v_1, \dots, v_n\}$ is contained in $\text{span}(D)$. It can therefore, be a basis for D .

We will choose the set $\{v_1, \dots, v_n\}$ as the basis set. Here are the reasons.

- These vectors minimize the average error due to projection
- These vectors maximize the variance

The above statements require more clarity and justification, which is given in the upcoming section.

5. PCA - Principal Component Analysis



The **red** points are data points. There are two lines, L_1 and L_2 .

Minimizing Average Projection Error

Let us ask the following question

- Which line represent the data points better? L_1 or L_2 ?

Visually, it is apparent that L_1 is better. The black line connecting a data point and L_1 represents the perpendicular distance between the data point and L_1 . Similarly, the blue line connecting a data point and L_2 represents the perpendicular distance between the data point and L_2 .

These lines represent the perpendicular distance. Which means, they also represent the error due to projection onto the corresponding lines.

The blue lines are longer than the black lines. Therefore, the average error due to projection onto L_1 is less than the average error due to projection onto L_2 .

Let's formalize this idea. Consider the following dataset with mean 0. Each data point is a column vector.

$$D = \{x_1, \dots, x_n\},$$

$$X = \begin{bmatrix} - & x_1^T & - \\ - & x_2^T & - \\ & \vdots & \\ - & x_n^T & - \end{bmatrix}$$

Every row in X is a data point.

A line is a span of a non-zero vector. Let's call it w . The projection of a data point x_i onto $\text{span}(w)$ is given by $\frac{(x_i^T w)w}{w^T w}$. If $\|w\| = 1$, then the projection is given by $(x_i^T w)w$. The error due to projection is given by the vector $x_i - (x_i^T w)w$.

Goal : Among all the lines, we would like to find the line such that the average projection error is minimum. If that line is represented as $\text{span}(\hat{w}_1)$, then

$$\hat{w}_1 = \arg \min_{w, \text{ s.t. } \|w\|=1} \left[\frac{1}{n} \sum_{i=1}^n \left\| x_i - (x_i^T w)w \right\|^2 \right]$$

Let us expand $\frac{1}{n} \sum_{i=1}^n \left\| x_i - (x_i^T w)w \right\|^2$

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n \left\| x_i - (x_i^T w)w \right\|^2 &= \frac{1}{n} \sum_{i=1}^n (x_i - (x_i^T w)w)^T (x_i - (x_i^T w)w) \\ &= \frac{1}{n} \sum_{i=1}^n x_i^T x_i - (x_i^T w)^2 - (x_i^T w)(w^T x_i) + (x_i^T w)^2 w^T w \\ &= \frac{1}{n} \sum_{i=1}^n x_i^T x_i - (x_i^T w)^2 - (x_i^T w)^2 + (x_i^T w)^2 \\ &= \frac{1}{n} \sum_{i=1}^n x_i^T x_i - (x_i^T w)^2 \end{aligned}$$

A few notes on the algebraic simplification :

- $w^T x_i = x_i^T w$
- $w^T w = \|w\|^2 = 1$

In $\frac{1}{n} \sum_{i=1}^n x_i^T x_i - (x_i^T w)^2$, the term $x_i^T x_i$ does not have w . Therefore, it doesn't influence the optimal $\hat{w}_1$.

$$\hat{w}_1 = \arg \min_{w, \text{ s.t. } \|w\|=1} \left[\frac{1}{n} \sum_{i=1}^n - (x_i^T w)^2 \right]$$

Instead of minimizing the objective function given above, we can maximize the following. Both will give us the same $\hat{w}_1$.

$$\hat{w}_1 = \arg \max_{w, st. ||w||=1} \left[\frac{1}{n} \sum_{i=1}^n (x_i^T w)^2 \right]$$

Let us expand $\frac{1}{n} \sum_{i=1}^n (x_i^T w)^2$

$$\begin{aligned} \frac{1}{n} \sum_{i=1}^n (x_i^T w)^2 &= \frac{1}{n} \sum_{i=1}^n (x_i^T w) (x_i^T w) \\ &= \frac{1}{n} \sum_{i=1}^n (w^T x_i) (x_i^T w) \\ &= \frac{1}{n} \sum_{i=1}^n w^T (x_i x_i^T) w \\ &= w^T \left(\frac{1}{n} \sum_{i=1}^n (x_i x_i^T) \right) w \\ &= w^T \left(\frac{1}{n} X^T X \right) w \\ &= w^T C w \end{aligned}$$

A few notes on the algebraic simplification :

- w^T and w were taken out of the summation, as they are common for all the data points.
- C is the covariance matrix as the mean of the data points is 0.

$\hat{w}_1$ is given as follows

$$\hat{w}_1 = \arg \max_{w, st. ||w||=1} [w^T C w] = \arg \max_{w, st. ||w||=1} \left[w^T \left(\frac{1}{n} X^T X \right) w \right]$$

Let $M = \frac{X}{\sqrt{n}}$. Then, $C = M^T M$.

$$\hat{w}_1 = \arg \max_{w, st. ||w||=1} [w^T M^T M w] = \arg \max_{w, st. ||w||=1} [||Mw||^2]$$

We have already solved a similar problem in section 2. $\hat{w}_1$ is the eigenvector of $M^T M = C$ corresponding to the largest eigenvalue of C .

The eigenvector of C corresponding to the largest eigenvalue of C is the vector which minimizes the average projection error.

Let's ask the same question, with slight variation :

- Among all the lines that are orthogonal to $\hat{w}_1$, which line minimizes the average projection error?

Formally, $\hat{w}_2$ is given as follows :

$$\hat{w}_2 = \arg \min_{\substack{w, \text{ st. } \\ ||w||=1 \\ \& \\ \hat{w}_1^T w=0}} \left[\frac{1}{n} \sum_{i=1}^n \left\| x_i - (x_i^T w) w \right\|^2 \right]$$

Once again, we can simplify this to the following:

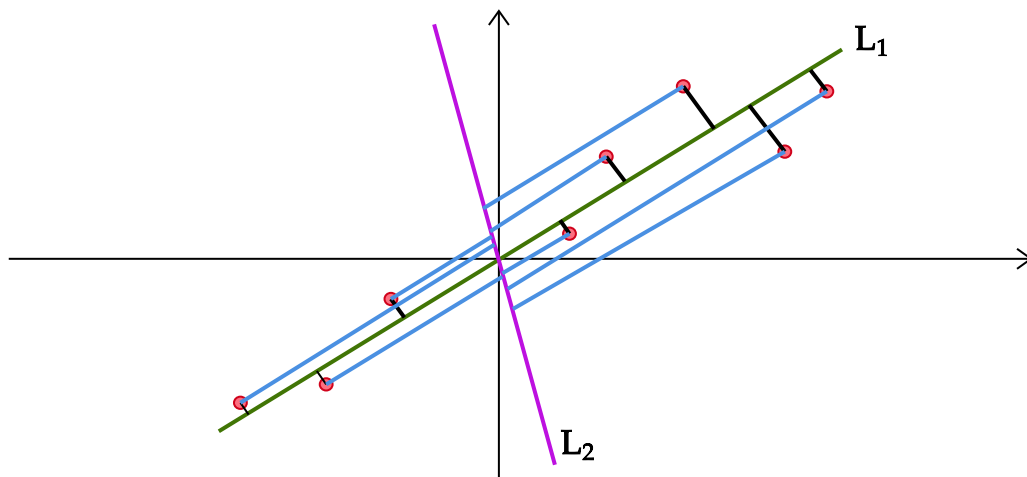
$$\hat{w}_2 = \arg \min_{\substack{w, \text{ st. } \\ ||w||=1 \\ \& \\ \hat{w}_1^T w=0}} \left[\frac{1}{n} \sum_{i=1}^n - (x_i^T w)^2 \right] = \arg \max_{\substack{w, \text{ st. } \\ ||w||=1 \\ \& \\ \hat{w}_1^T w=0}} [w^T M^T M w] = \arg \max_{\substack{w, \text{ st. } \\ ||w||=1 \\ \& \\ \hat{w}_1^T w=0}} [||Mw||^2]$$

A similar problem was solved in section 2. $\hat{w}_2$ is the eigenvector of $M^T M = C$, corresponding to the second largest eigenvalue of C .

If $\text{rank}(X) = k$, then $\text{rank}(C) = k$. We can ask the same questions k times.

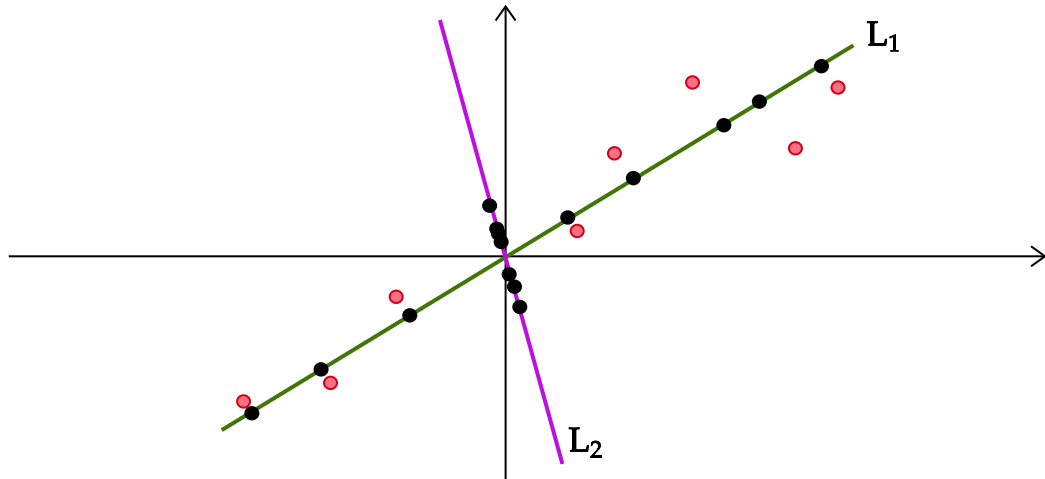
We will get k eigenvectors corresponding to the top k eigenvalues of C . These eigenvalues will be greater than 0. These eigenvectors are referred to as the **principal components**.

Maximizing Variance



Lets revist this image. We have already seen that L_1 is better than L_2 , as the average error due to projection onto L_1 is less than the average error due to projection onto L_2 .

Lets shift the focus from the **average error due to projection** to the **projections** themselves.



The black points are the projections. The projections onto L_1 are widely "spread" across L_1 , whereas, the projections onto L_2 are narrowly "spread" across L_2 . The data points are more oriented along direction of L_1 when compared to L_2 .

More the spread, better the line. How do we quantify "spread"? Answer : **Variance**. Each line can be expressed as the span of a unit vector. The projections onto that line is a scalar multiple of the unit vector. The norm of a unit vector is 1. Therefore, the scalar that multiplies the unit vector represents the length.

Let w represent a unit vector. The projection of a data point x_i onto $span(w)$ is given by $(x_i^T w)w$. Here, $(x_i^T w)$ is the scalar. The line that best represents the data points shall be the line along which the variance of the projection is maximum. Let us denote the best line as $span(\hat{w})$. Then,

$$\hat{w} = \arg \max_{w, st. ||w||=1} \mathbf{Var}\left((x_1^T w), (x_2^T w), \dots, (x_n^T w)\right)$$

To find the variance, we would require the mean.

$$\mathbf{Mean}\left((x_1^T w), (x_2^T w), \dots, (x_n^T w)\right) = \frac{1}{n} \sum_{i=1}^n (x_i^T w)$$

$$\begin{aligned}
&= \left(\frac{1}{n} \sum_{i=1}^n x_i^T \right) w \\
&= (0)w \\
&= 0
\end{aligned}$$

The mean of the data points were assumed to be 0. Therefore, $\frac{1}{n} \sum_{i=1}^n x_i^T = 0$

$\hat{w}$ is given as follows.

$$\hat{w} = \arg \max_{w, st. ||w||=1} \left[\frac{1}{n} \sum_{i=1}^n (x_i^T w - 0)^2 \right] = \arg \max_{w, st. ||w||=1} \left[\frac{1}{n} \sum_{i=1}^n (x_i^T w)^2 \right]$$

Let's simplify $\frac{1}{n} \sum_{i=1}^n (x_i^T w)^2$

$$\begin{aligned}
\frac{1}{n} \sum_{i=1}^n (x_i^T w)^2 &= \frac{1}{n} \sum_{i=1}^n (x_i^T w) (x_i^T w) \\
&= \frac{1}{n} \sum_{i=1}^n (w^T x_i) (x_i^T w) \\
&= w^T \left(\frac{1}{n} \sum_{i=1}^n x_i x_i^T \right) w \\
&= w^T C w
\end{aligned}$$

Therefore,

$$\hat{w} = \arg \max_{w, st. ||w||=1} [w^T C w]$$

We have already sloved this problem while minimizing the average error due to projection. The key insight is that, the line that minimizes the average projection error is the line that maximizes the variance.

Let w_i be the eigenvector of C corresponding to the i^{th} largest eigenvalue (λ_i) of C . Then,

$$\begin{aligned}
w_i^T C w_i &= w_i^T (\lambda_i w_i) \\
&= \lambda_i (w_i^T w_i) \\
&= \lambda_i
\end{aligned}$$

$w_i^T C w_i$, which denotes the variance along w_i is simplified to λ_i . This implies that the variance along w_i is its corresponding eigenvalue.

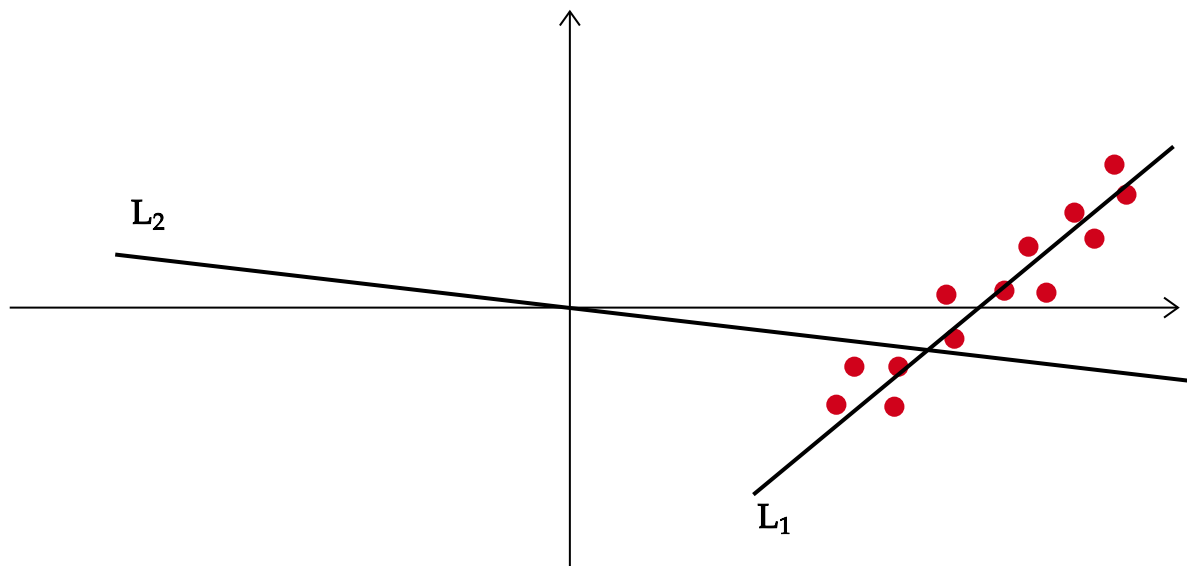
Summary

Let C be the covariance matrix. $w_1, \dots, w_k$ are the eigenvectors of C with $\lambda_1, \dots, \lambda_k$ as the corresponding eigenvalues. Here, $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_k$. Then,

- w_1 is the direction along which the variance of the projected data is maximum, and equivalently, it is also the direction along which the average projection error is minimum. The variance along w_1 is λ_1 .
 - Among all the directions orthogonal to w_1 , w_2 is the direction along which the variance of the projected data is maximum, and equivalently, it is also the direction along which the average projection error is minimum. The variance along w_2 is λ_2 .
 - Among all the directions orthogonal to $\{w_1, \dots, w_{k-1}\}$, w_k is the direction along which the variance of the projected data is maximum, and equivalently, it is also the direction along which the average projection error is minimum. The variance along w_k is λ_k .
-

6. Mean-Centering the Dataset

So far, we have assumed that the mean of the data points is 0. In general, this may or may not be true.



Structurally, the data points are aligned along L_1 . However, the line that minimizes the average projection error would look something similar to L_2 . This is because, the line that minimizes the average projection error must pass through the origin.

Here is a solution. First, we find the mean of the data points, Let's denote it as $\bar{x}$.

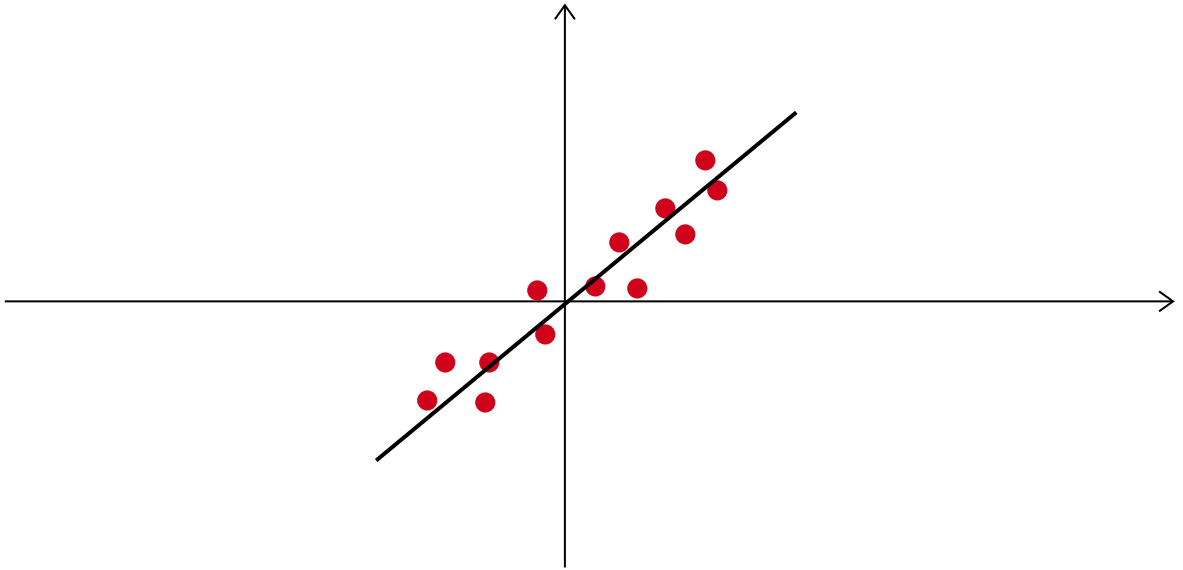
$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

From every data point, we subtract the mean.

$$x_i = x_i - \bar{x}$$

Now, the mean of the data points will be 0. This process is known as "mean-centering" the dataset.

After mean-centering the dataset shown in the above figure, the dataset would look as follows.



The line that minimizes the average error due to projection would also be the line that captures the structural alignment of the data points.

7. PCA in Higher Dimensions

Suppose, we have n data points in $\mathbb{R}^d$. If $d > n$, computing $C = \frac{1}{n} X^T X$, which is a $d \times d$ matrix, and finding its eigenvalues will be computationally costly. In this section, we shall look at a workaround for this.

FACT : The non-zero eigenvalues of $X^T X$ and XX^T are same.

XX^T is an $n \times n$ matrix.

Let $\lambda_1, \dots, \lambda_k$ be the k non zero eigenvalues of $C = \frac{1}{n} X^T X$. Their corresponding eigenvectors are $w_1, \dots, w_k$, where $\|w_i\| = 1, \forall i \in \{1, \dots, k\}$. Here, $\lambda_1 \geq \dots \geq \lambda_k$

Let $l_1, \dots, l_k$ be the k non zero eigenvalues of XX^T . Their corresponding eigenvectors are $\beta_1, \dots, \beta_k$, where $||\beta_i|| = 1, \forall i \in \{1, \dots, k\}$. Here, $l_1 \geq \dots \geq l_k$

Then,

$$\begin{aligned}\frac{1}{n}X^T X w_i &= \lambda_i w_i \\ X^T X w_i &= (n\lambda_i)w_i \\ XX^T(Xw_i) &= (n\lambda_i)Xw_i\end{aligned}$$

Clearly, $l_i = n\lambda_i$. Therefore, $\frac{l_i}{n} = \lambda_i$.

Let us find $||Xw_i||^2$

$$\begin{aligned}||Xw_i||^2 &= (Xw_i)^T(Xw_i) \\ &= w_i^T X^T X w_i \\ &= w_i^T (n\lambda_i)w_i \\ &= (n\lambda_i)w_i^T w_i \\ &= n\lambda_i\end{aligned}$$

$$\beta_i = \frac{Xw_i}{||Xw_i||} = \frac{Xw_i}{\sqrt{n\lambda_i}} = \frac{Xw_i}{\sqrt{l_i}}$$

$$\begin{aligned}\beta_i &= \frac{Xw_i}{\sqrt{l_i}} \\ \sqrt{l_i}\beta_i &= Xw_i \\ \frac{\sqrt{l_i}}{n}X^T\beta_i &= \frac{1}{n}X^T X w_i \\ \frac{\sqrt{l_i}}{n}X^T\beta_i &= \lambda_i w_i \\ \frac{\sqrt{l_i}}{n\lambda_i}X^T\beta_i &= w_i \\ \frac{\sqrt{l_i}}{l_i}X^T\beta_i &= w_i \\ \frac{X^T\beta_i}{\sqrt{l_i}} &= w_i\end{aligned}$$

We have $\frac{X^T \beta_i}{\sqrt{l_i}} = w_i$ and $\frac{l_i}{n} = \lambda_i$.

PCA Steps when $d > n$

- Mean Center the dataset
 - Each data point is a row in matrix X
 - Find XX^T
 - The k non-zero eigenvalues of XX^T are $l_1, \dots, l_k$. Here, $l_1 \geq \dots \geq l_k$
 - β_i is the eigenvector of XX^T , that corresponds to eigenvalue l_i . Here, $\|\beta_i\| = 1$
 - The eigenvalues of $C = \frac{1}{n}X^T X$ are $\lambda_1, \dots, \lambda_k$. Here, $\lambda_i = \frac{l_i}{n}$
 - w_i is the eigenvector of C , that corresponds to λ_i . Here, $w_i = \frac{x^T \beta_i}{\sqrt{l_i}}$. Here, $\|w_i\| = 1$
-